

Marketing Analytics Datathon

Step-by-Step Project Process

How to run the Python pipeline, review outputs in Excel, and explain the analysis.

- ? Python is the source of truth for cleaning, joins, attribution, ML, dashboard generation, slide generation, and validation.
- ? Excel is used for reviewing generated reports, building PivotTables, checking numbers, and supporting the story.
- ? Follow the steps in order. Each step has a goal, actions, and output/checkpoint.

Final dashboard: `public/index.html`

Final deck: `outputs/presentation/Marketing_Analytics_Executive_Deck_v4.pptx`

Validator: `analytics_case_study/06_validate_metrics.py`

Step 1 - Understand The Workflow

Goal

Know which parts are done in Python and which parts are done in Excel.

Actions

- ? Use Python as the source of truth for cleaning, integration, attribution, machine learning, dashboard generation, slide generation, and validation.
- ? Use Excel to inspect outputs, build PivotTables, verify business logic, and explain results.
- ? Do not manually rebuild the entire project in Excel. Some steps can be reviewed in Excel, but should be produced by Python.

Output / Checkpoint

- ? Clear split: Python produces reliable outputs; Excel is used for review and presentation support.

Step 2 - Install And Confirm Python Dependencies

Goal

Make sure the project can run end to end.

Actions

- ? Open PowerShell in the project folder.
- ? Use the installed Python interpreter:
C:\Users\andre\AppData\Local\Programs\Python\Python312\python.exe
- ? Run: `python -m pip install -r requirements.txt`, or use the full interpreter path if python is not on PATH.
- ? Confirm dependencies install without fatal errors.

Output / Checkpoint

- ? Python environment ready to run the pipeline.

Step 3 - Run Data Cleaning In Python

Goal

Convert raw Excel exports into clean, consistent tables.

Actions

- ? Run: analytics_case_study/01_data_cleaning.py.
- ? The script reads all raw files from Analytics Case Study/.
- ? It cleans nulls, dates, numeric fields, domains, countries, industries, boolean flags, and marketing channel categories.
- ? It deduplicates opportunities by _opportunity_id and accounts by accountid.
- ? It writes cleaned parquet files into data/cleaned/.

Output / Checkpoint

- ? data/cleaned/opportunities.parquet
- ? data/cleaned/accounts.parquet
- ? data/cleaned/email_engagements.parquet
- ? data/cleaned/web_engagements.parquet
- ? data/cleaned/ad_metrics.parquet
- ? data/cleaned/6sense_campaign.parquet
- ? data/cleaned/6sense_segments.parquet
- ? data/cleaned/icp_database.parquet

Step 4 - Review Cleaned Data In Excel

Goal

Use Excel for sanity checks, not as the main cleaning tool.

Actions

- ? Open the raw source files or generated Excel reports if needed.
- ? Check that opportunity IDs are unique after cleaning.
- ? Check that account IDs are unique after cleaning.
- ? Check that amount fields are numeric and won flags are TRUE/FALSE.
- ? Check that channel_category values are understandable: 6sense_display, email_mqa, web_inbound, event, webinar, referral, existing_client, other, etc.

Output / Checkpoint

- ? Confidence that cleaned tables are usable before analysis.

Step 5 - Build Integrated Tables In Python

Goal

Join cleaned data into account-level and analysis-ready tables.

Actions

- ? Run: analytics_case_study/02_data_integration.py.
- ? The script builds master_account by joining account data with segments, campaign activity, email activity, web activity, opportunities, and ICP contacts.
- ? It builds channel_pipeline for pipeline, won revenue, win rate, spend, ROI, and cost per opportunity.
- ? It builds funnel_metrics for impressions, clicks, form fills, email engagement, opportunities, and won deals.
- ? It builds creative_performance for ad and creative-level performance.

Output / Checkpoint

- ? data/integrated/master_account.parquet
- ? data/integrated/channel_pipeline.parquet
- ? data/integrated/funnel_metrics.parquet
- ? data/integrated/creative_performance.parquet

Step 6 - Generate Basic Excel Analysis Reports

Goal

Create Excel workbooks that can be opened, reviewed, and used for PivotTables.

Actions

- ? Run: `analytics_case_study/03_analysis.py`.
- ? Open `outputs/analysis/channel_roi.xlsx` to review channel spend, pipeline, won revenue, ROI, and cost per opportunity.
- ? Open `outputs/analysis/segment_conversion.xlsx` to review segment, industry, and seniority performance.
- ? Open `outputs/analysis/creative_performance.xlsx` to review ad CTR, CPC, CPM, and creative attributes.
- ? Open `outputs/analysis/email_campaign_performance.xlsx` to review email performance by campaign, seniority, industry, and quarter.
- ? Open `outputs/analysis/budget_recommendation.xlsx` to review budget scenarios.

Output / Checkpoint

- ? Excel reports in `outputs/analysis/`.

Step 7 - Build Or Review Excel PivotTables

Goal

Use Excel where it is strongest: summarizing and explaining the outputs.

Actions

- ? Create PivotTables from the generated Excel reports.
- ? Pivot channel ROI by Channel with Deals, Total Pipeline, Won Revenue, Win Rate, Spend, Pipeline ROI, and Revenue ROI.
- ? Pivot segment conversion by Segment with Deals, Won Deals, Pipeline, Average Deal, and Win Rate.
- ? Pivot email performance by Seniority and Campaign.
- ? Pivot creative performance by Platform, Copy Tone, Asset Type, CTA Type, and Ad Name.
- ? Use slicers for channel, segment, industry, quarter, and profile fit if building an Excel dashboard.

Output / Checkpoint

- ? A review workbook or optional Excel dashboard for manual exploration.

Step 8 - Run Attribution In Python

Goal

Calculate sourced, influenced, first-touch, last-touch, linear, and time-decay attribution.

Actions

- ? Run: analytics_case_study/03b_attribution.py.
- ? Python builds a touchpoint table from 6sense, email, and web activity.
- ? It resolves opportunity domains through accounts when needed.
- ? It links touchpoints to opportunities if they happened within 365 days before opportunity creation.
- ? It applies six attribution models and writes the results.

Output / Checkpoint

- ? outputs/analysis/attribution_models.xlsx
- ? data/integrated/attribution_results.parquet

Step 9 - Review Attribution In Excel

Goal

Explain attribution results without trying to manually rebuild them.

Actions

- ? Open outputs/analysis/attribution_models.xlsx.
- ? Compare Marketing Sourced vs Marketing Influenced.
- ? Compare First-Touch vs Last-Touch to identify awareness channels vs closing/support channels.
- ? Use Linear as a fair shared-credit view.
- ? Use Time-Decay as the most budget-relevant directional view because it rewards recent touches more.

Output / Checkpoint

- ? Attribution explanation ready for dashboard/deck.

Step 10 - Run Advanced Analytics In Python

Goal

Build analyses that are difficult or unrealistic to maintain in Excel.

Actions

- ? Run: analytics_case_study/03c_advanced_analytics.py.
- ? Python trains the Random Forest win probability model and scores open deals.
- ? It builds account coverage analysis: Not Reached, Email Only, 6sense Only, Both Channels.
- ? It calculates deal velocity by channel and segment.
- ? It builds journey sequences such as email_mqa -> 6sense_display.
- ? It builds 6QA/profile fit performance, targeting priority matrix, and cohort trends.

Output / Checkpoint

- ? outputs/analysis/advanced_analytics.xlsx
- ? data/integrated/win_probability.parquet
- ? data/integrated/account_coverage.parquet
- ? data/integrated/deal_velocity.parquet
- ? data/integrated/journey_sequences.parquet
- ? data/integrated/targeting_matrix.parquet
- ? data/integrated/cohort_analysis.parquet

Step 11 - Review Advanced Analytics In Excel

Goal

Use Excel to inspect and explain advanced outputs.

Actions

- ? Open outputs/analysis/advanced_analytics.xlsx.
- ? Review feature importance and win probability scores, but remember Excel did not train the model.
- ? Review account coverage to identify unreached target accounts.
- ? Review deal velocity to understand which channels close faster.
- ? Review journey sequences to understand common paths before won deals.
- ? Review cohort analysis to explain whether pipeline quality is improving or declining over time.

Output / Checkpoint

- ? Advanced insights ready for final story.

Step 12 - Generate The Dashboard In Python

Goal

Create the final interactive dashboard.

Actions

- ? Run: analytics_case_study/04_html_dashboard.py.
- ? The script reads integrated outputs and renders charts into a self-contained HTML dashboard.
- ? It saves the dashboard to outputs/dashboard/Marketing_Analytics_Dashboard.html.
- ? It copies the same file to public/index.html for static deployment.
- ? Open public/index.html in a browser to review the dashboard.

Output / Checkpoint

- ? outputs/dashboard/Marketing_Analytics_Dashboard.html
- ? public/index.html

Step 13 - Generate The PowerPoint Deck In Python

Goal

Create an executive-ready presentation from the analysis outputs.

Actions

- ? Run: `analytics_case_study/05_presentation.py`.
- ? The script builds a 21-slide deck with methodology, attribution, funnel, channel performance, creative/email, advanced analytics, and recommendations.
- ? Open the generated PPTX and make final wording or formatting edits if needed.

Output / Checkpoint

- ? `outputs/presentation/Marketing_Analytics_Executive_Deck_v4.pptx`

Step 14 - Validate Everything

Goal

Confirm the project is internally consistent before submission.

Actions

- ? Run: analytics_case_study/06_validate_metrics.py.
- ? Confirm validation passes with 0 errors.
- ? Confirm public/index.html is synced with
outputs/dashboard/Marketing_Analytics_Dashboard.html.
- ? Confirm key Excel reports exist in outputs/analysis/.
- ? Confirm the final deck exists and opens.

Output / Checkpoint

- ? A validated project ready to submit or present.

Step 15 - Final Business Story

Goal

Explain the work clearly to judges or stakeholders.

Actions

- ? Start with the business question: which marketing activities contribute to pipeline and revenue?
- ? Explain sourced pipeline vs influenced pipeline.
- ? Show attribution models to explain channel roles.
- ? Show account coverage to prove the largest growth lever is reaching more target accounts.
- ? Show cohort analysis to warn that pipeline growth must be balanced against win-rate quality.
- ? End with recommendations: reach strong-fit unreached accounts, start with email, test 6sense overlay, protect win rate, and report sourced plus influenced metrics together.

Output / Checkpoint

- ? Clear executive narrative for dashboard and deck.